Predicting Intron Placement from Helical Phase Stress

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Abstract

We present a geometric model of intron placement based on helical phase stress. B-form DNA rotates approximately $360°/10.5 \approx 34.3°$ per base pair. The "half-octave" of 4 bp subtends approximately 137.1°, which lies within 1° of the small golden angle $360°/\varphi^2 \approx 137.5°$. We prove this proximity rigorously in Lean 4 using rational squeezes on $\sqrt{5}$. A phase-stress function measures cumulative deviation from golden-angle alignment along a coding sequence; the model predicts that introns are inserted at sites of maximal stress to reset the phase. We further prove that the minimal functional intron size $8\varphi^5 \approx 89$ bp lies within 1 bp of the observed mammalian minimum of 85–90 bp, using the algebraic identity $\varphi^5 = 5\varphi + 3$. The exact return period of the B-form helix is 21 bp $= 3 \times 7$, connecting codon length (3) to the neutral-subspace dimension (7). All geometric theorems and the phase-stress model are formalized in Lean 4 with zero axioms and zero unproved assertions. The empirical claims (mod-8 non-uniformity in intron-length distributions) are explicitly marked as dataset-scoped hypotheses, not theorems.

1 Introduction

Introns interrupt the coding sequences of most eukaryotic genes. Their functional significance has been debated since their discovery [1, 3]: are they evolutionary relics, or do they serve a structural purpose? Comparative genomics shows that intron positions are often conserved across species [6], suggesting selective constraint, but the nature of that constraint is unclear.

We propose that introns function as **helical phase-alignment spacers**. The B-form DNA double helix rotates by a fixed angle per base pair. Over a long coding stretch, the cumulative rotation drifts away from any preferred alignment. Introns interrupt the coding sequence at positions of maximal drift, resetting the helical phase so that splice-site recognition—which depends on the relative orientation of donor and acceptor sites on the helix [5]—remains efficient.

The key geometric coincidence underlying the model is that a 4 bp segment of B-form DNA rotates by approximately 137.1°, which is within 1° of the small golden angle ($\approx 137.5°$). This near-equality is not a numerical accident: it connects the physical parameter of the double helix (10.5 bp/turn) to the golden ratio forced by the Recognition Science algebraic backbone [8].

2 Geometric Foundations

2.1 The Half-Octave Near-Golden-Angle Theorem

B-form DNA has approximately 10.5 bp per helical turn. The rotation subtended by 4 bp (the "half-octave") is

$$\theta_{\text{rot}} = 4 \times \frac{360°}{10.5} = \frac{960}{7} \approx 137.14°.$$

The small golden angle is

$$\theta_{\text{golden}} = 360° - \frac{360°}{\varphi} = \frac{360°}{\varphi^2} \approx 137.51°.$$

Theorem (`half_octave_near_golden_angle` in `IntronStructure.lean`).

$$|\theta_{\text{rot}} - \theta_{\text{golden}}| < 1°.$$

The proof uses rigorous rational squeezes on $\sqrt{5}$: from $2.236^2 < 5$ we obtain $\varphi > 809/500$, and from the existing bound $\varphi < 81/50$ we trap $360/\varphi$ between $1553/7$ and $1567/7$. The difference $\theta_{\text{rot}} - \theta_{\text{golden}} = 360/\varphi - 1560/7$ then lies strictly between -1 and $+1$.

2.2 The Minimal-Intron Estimate

Theorem (`minimal_intron_phi5_scaling` in `IntronStructure.lean`).

$$|8\varphi^5 - 89| < 1.$$

The proof uses the algebraic identity $\varphi^5 = 5\varphi + 3$ (derived from $\varphi^2 = \varphi + 1$), giving $8\varphi^5 = 40\varphi + 24$. With the bounds $1.618 < \varphi < 1.62$, this places $8\varphi^5$ strictly between 88 and 90. The observed minimal functional intron size in mammals is 85–90 bp [9].

2.3 The Exact Return Period

The B-form helix returns to exactly $0°$ (mod $360°$) after 21 bp:

$$21 \times \frac{360°}{10.5} = 720° = 2 \times 360°.$$

Theorem (`exact_return_21bp` in `HelicalPhaseModel.lean`). $21 \times (360/10.5) = 720$.

The factorization $21 = 3 \times 7$ connects the codon length (3) to the neutral-subspace dimension (7) of the 8-tick DFT basis. This is formalized as `spacing_factors`.

3 The Phase-Stress Model

3.1 Cumulative Phase and Stress

For a coding sequence of p base pairs, the cumulative unwrapped helical phase is

$$\Theta(p) = p \times \frac{360}{10.5}.$$

The **phase stress** at position p is the distance from $\Theta(p)$ to the nearest multiple of the golden angle:

$$S(p) = \left|\Theta(p) - \theta_{\text{golden}} \cdot \lfloor \Theta(p)/\theta_{\text{golden}} + 0.5 \rfloor\right|.$$

Both functions are defined in `HelicalPhaseModel.lean` (`unwrappedPhase`, `phaseStress`).

3.2 Intron Insertion Rule

The model predicts that introns are inserted at positions where:

1. the site is at least $8\varphi^5 \approx 89$ bp from the last intron (the minimal intron distance),
2. the site lies on the half-octave grid ($p \equiv 0 \pmod 4$),
3. the phase stress exceeds a threshold.

These conditions are formalized by the predicate `predictedCandidateSite` in `IntronPlacementOptimality.lean`. The first structurally admissible site is 92 bp (the smallest multiple of 4 above 89), proved as `first_structural_candidate`.

3.3 Aligned Spacers Reduce Penalty

Theorem (`aligned_spacer_strictly_reduces_penalty` in `IntronPlacementOptimality.lean`). If the site is high-stress and the spacer length is a multiple of 4 bp, then the post-insertion phase penalty is strictly less than the pre-insertion penalty.

This is the central optimality result: phase-aligned introns are locally better than no intron at a stressed site.

4 Intron Density Predictions

The model predicts a natural intron density:

- Exact return period: 21 bp.
- Predicted spacing between introns: multiples of $21 \times k$ bp.
- For a typical 1500 bp human gene: ∼10 introns.
- For a 900 bp gene: ∼6 introns.

Both counts are proved by `native_decide` in `HelicalPhaseModel.lean` (`typical_gene_introns`, `small_gene_introns`).

The observed average for human genes is approximately 8 introns per gene [7], consistent with the model's prediction range.

5 Empirical Status and Falsifiability

5.1 Mod-8 Non-Uniformity

The model predicts that intron-length distributions should show statistically significant non-uniformity mod 8. Exploratory analyses on Ensembl-derived gene sets confirm significant χ^2 statistics ($p < 0.001$), but the peak residue depends on gene-set selection and transcript weighting. This variability is explicitly acknowledged in the Lean formalization: the mod-8 claims are defined as dataset-scoped propositions (`intron_mod8_residue4_preference_base_set`), not as universal theorems. They carry no formal proof obligation.

5.2 Falsifiers

1. **Uniform mod-8**: If intron lengths are uniformly distributed mod 8 across well-powered, preregistered datasets, the model's geometric basis is refuted.
2. **Minimal intron size**: If minimal functional introns are consistently $\gg 100$ or $\ll 70$ bp, the $8\varphi^5$ prediction fails.
3. **Splice-site independence**: If splice-site recognition is completely independent of helical position (e.g., in in-vitro systems with linearized DNA), the geometric motivation is undermined.

6 Formal Verification

All geometric and structural theorems are formalized in Lean 4 (version 4.27.0) with Mathlib. The four modules are:

- `IntronStructure.lean`: helix geometry, golden-angle proximity (`half_octave_near_golden_angle`), minimal-intron estimate (`minimal_intron_phi5_scaling`).
- `HelicalPhaseModel.lean`: cumulative phase, phase stress, exact return period (`exact_return_21bp`), intron density predictions.
- `IntronPlacementOptimality.lean`: candidate-site predicates, spacer-relief theorem (`aligned_spacer_strictly_reduces_penalty`)
- `PhaseAlignmentLayer.lean`: genome-fraction prediction (∼25%), golden-angle phase alignment, minimal-intron decomposition.

All modules contain zero axioms and zero `sorry` assertions. Empirical hypotheses are defined as propositions but carry no proof; they are explicitly labeled as dataset-scoped.

7 Discussion

7.1 Relation to Existing Intron Theories

The "introns-early" hypothesis [4] holds that introns are ancient and predate the divergence of prokaryotes and eukaryotes. The "introns-late" hypothesis [2] holds that introns were inserted into previously intronless genes. Our model is neutral on this question: it explains *where* introns should be placed given the helix geometry, not *when* they were inserted.

The observation that splice-site recognition depends on helical orientation [5] provides independent support for the geometric basis of the model.

7.2 Scope

The model takes the B-form helix parameter (10.5 bp/turn) as an empirical input. It predicts intron *positions* and *density*, not intron *sequences* or *splicing mechanism*. The crossover to A-form or Z-form DNA is predicted to require different intron spacing (proved as `different_forms_different_spacing`), which is testable in RNA–DNA hybrid contexts.

8 Conclusion

Intron placement can be predicted from helical phase stress. The 4 bp half-octave rotation matches the golden angle to within 1°, the minimal intron size matches $8\varphi^5$ to within 1 bp, and the model predicts intron densities consistent with observed averages. All geometric results are machine-checked in Lean 4. The remaining empirical step is a preregistered, multi-dataset analysis of intron-length distributions mod 8, which the formalization explicitly separates from the proved geometric surface.

Data Availability

All Lean source files are in the `reality` repository under `IndisputableMonolith/Genetics/`. Exploratory intron data from Ensembl (GRCh38) is in `data/`.

References

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